General Elections and Housing Transaction Volumes

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Abstract

Using state-level monthly pending home sales from 2012 to 2025, we examine whether general elections generate measurable distortions in housing transaction volumes. We employ a interaction design that decomposes housing dynamics into a non-election seasonal baseline, a presidential election excess, and a midterm election excess - using non-election trends as an explicit counterfactual. Presidential elections suppress pending sales by approximately 3 to 5 percent below the seasonal baseline in the five months preceding the vote, consistent with predictions made by Bernanke (1983). However, upon resolution of uncertainty, sales cease to be suppressed rather than exhibiting a pent-up demand spike. Midterm elections produce no comparable deviation. These findings are significant under preferred clustering specification but lose significance under conservative cycle-level inference, reflecting the fundamental constraint of observing only three full presidential cycles.

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1 Introduction

1.1 Background and Literature Review

Election seasons are hard to ignore. For months on end, competing visions of tax policy, regulation, and federal spending dominate news cycles, dinner table conversations, and household financial planning. The uncertainty is palpable: will mortgage interest deductions survive the next administration? Will interest rates shift in response to a new fiscal agenda? For families in the market for housing, these questions become particularly salient - a home purchase is a financial commitment that is notoriously difficult to reverse once pen is put to paper. In the face of uncertainty, sometimes the best course of action is no action at all.

For as long as this nugget of wisdom has existed, economists have attempted to theoretically formalize and empirically examine it. In his seminal work on irreversibility and investment timing, Ben Bernanke (1983) showed that when an investment is costly to reverse and new information is expected to arrive, the value of waiting can even exceed the potential returns by investing early. For those who can wait (or those with high loss aversion), this framework predicts reductions in - or even total cessation of - investment until new information is made available. Once uncertainty resolves, a temporary surge that ultimately returns to baseline is expected, as patient investors return into the market - and as the market normalizes.

Subsequent empirical work has supported the depressive effect of uncertainty across various economic actors, contexts, and study designs. Specifically, Coibion et al. (2024) used a randomized information treatment to show that higher perceived uncertainty causes households to reduce spending, with particularly robust effects on purchases of durable goods. Aaberge et al. (2017) found that higher measures of uncertainty in China increase precautionary savings, driven primarily by reductions in semi-durable and durable consumption.

Elections, due to their recurrent nature and a pre-defined resolution of uncertainty once results are announced, also have a prominent position in the literature. Canes-Wrone and

Park (2014) examined the link between elections, irreversible investments, suggesting that uncertainty surrounding elections is not only associated with slowdowns in investment behavior across sectors, but also with housing market slowdowns at the state level. Focusing on the latter, two studies provide the closest precedent for our analysis. Nguyen and Vergara-Alert (2023) use CoreLogic transaction records and state-level annual data from 1982 to 2018 to document that gubernatorial elections reduce housing transaction volumes by approximately 2,000 to 3,200 transactions per state-year. The effect scales with electoral uncertainty - close races and elections with absent incumbents produce larger declines - and reverses in the two years following the election, consistent with pent-up demand being released once uncertainty resolves. Choi (2023) examines the same question at the county level, finding that a unit increase in state political uncertainty is associated with approximately a 3.4 percentage point decrease in local house price growth, with the impact weaker in areas with higher homeownership rates. Christidou and Fountas (2017) also document that uncertainty affects US state-level housing markets, though their analysis centers on price volatility using time-series methods rather than on transaction volume.

Two gaps in this literature motivate our analysis. First, existing work focuses almost exclusively on gubernatorial elections, where policy transmission is immediate and geographically targeted, and where staggered election timing across states provides the cross-sectional variation that enables strong causal identification. General elections, however, have received comparatively little attention in this context - despite occurring on a fixed, predictable schedule and carrying a longer transition period (approximately eleven weeks) during which policy direction remains ambiguous. Second, the existing studies rely on annual data, which would not be able to capture within-year timing dynamics at a high resolution - which is central to Bernanke's theoretical predictions.

1.2 This Paper

This paper asks: do general elections (presidential and the midterms) generate measurable distortions in state-level housing transaction volumes, and if so, does the timing of these distortions follow the predictions of standard uncertainty theory? We employ a panel event study framework that decomposes housing transaction patterns into a non-election seasonal baseline, a presidential election excess, and a midterm election excess, providing an explicit counterfactual - the seasonal pattern in non-election years - against which election-year deviations can be measured.

Using state-level monthly data on pending home sales from 2012 to 2025, we find a pattern partially consistent with Bernanke's (1983) option value framework: pending sales dip below the seasonal norm in the months leading up to presidential elections, then revert toward baseline after the January inauguration without a predicted spurt of pent-up demand. This pattern is specific to presidential cycles; midterm elections show no comparable deviation. That said, the small number of available presidential cycles (three, after excluding 2020 due to pandemic-era confounds) and the sensitivity of our results to clustering assumptions signals caution in generalizing these findings.

The remainder of the paper proceeds as follows. Section 2 describes our data sources and sample construction. Section 3 presents our methodology, building up our identification strategy step by step. Section 4 reports results, and Section 5 discusses limitations and concludes.

2 Data and Descriptive Statistics

2.1 Housing Data

We obtain housing market data from publicly available state-level monthly estimates from the Redfin Data Center (Redfin, 2025). Redfin aggregates Multiple Listing Service (MLS) feeds to capture housing market fluctuations across all 50 states and the District of Columbia. For our analysis, we use aggregate pending home sales as our outcome variable of interest. Per Redfin (2025), pending home sales are "the total number of homes that went under contract," representing the number of sellers who have accepted the terms and conditions of an offer to purchase a property. This offers a more immediate proxy for buyer decision-making than closed sales, which reflect economic decisions made in prior periods due to financing, inspection, and closing timelines.

We apply a natural log transformation to pending home sales prior to analysis. Housing volume varies substantially with state population, producing a right-skewed distribution; the log transformation compresses this variation and allows regression coefficients to be interpreted as approximate percentage changes in sales volume, instead of raw counts seen in other analyses.

It is important to note that Redfin data is reported raw and not seasonally adjusted. This is advantageous for our purposes: seasonal adjustment algorithms can be somewhat opaque in their estimations, and their application could inadvertently absorb some of the election-cycle variation we aim to identify. Instead, we control for seasonality explicitly within our model specification, giving us full transparency over how seasonal patterns are handled.

Redfin data is available from January 2012 onward. This is the binding constraint on our sample, restricting the analysis to the 2012-2025 period and providing coverage of four presidential election cycles (2012, 2016, 2020, 2024) and three midterm cycles (2014, 2018, 2022).

2.2 Macroeconomic Controls and Weights

We supplement this housing dataset with four additional macroeconomic variables drawn from the following sources:

2.2.1 Mortgage Rates

Interest rates are the primary macroeconomic determinant of housing affordability: monthly mortgage payments are directly linked to borrowing costs, and shifts in the rate environment can push marginal buyers in or out of the market. In a panel that spans Federal Reserve tightening and easing cycles, failing to account for rate movements risks attributing interest-rate-driven fluctuations to election effects. We use the Freddie Mac Primary Mortgage Market Survey (PMMS), which reports the weekly national average for 30-year fixed-rate mortgages from 1971 to present (Freddie Mac, n.d.). Weekly observations are averaged to the monthly level for consistency with our panel structure. The PMMS is a national series, which means that the same mortgage rate would apply to all states in a given month, even though actual mortgage rates might vary from state to state, or even from one home purchase to the next.

2.2.2 State Unemployment Rates

Local labor market conditions affect both household income expectations and the willingness to commit to large, illiquid purchases. A state experiencing a localized employment shock - a plant closure, a seasonal downturn in a dominant industry, or the effects of a pandemic - may see reduced housing demand for reasons entirely unrelated to the electoral calendar. Monthly state-level unemployment rates are obtained from the Bureau of Labor Statistics' Local Area Unemployment Statistics (LAUS) program, accessed via the FRED API (Bureau of Labor Statistics, n.d.). These are reported monthly at the state level, matching our panel structure directly.

2.2.3 State GDP Growth

Broader economic momentum - whether a state's economy is accelerating or decelerating - plausibly influences housing demand. We use quarterly state GDP data from the Bureau of Economic Analysis's State Quarterly GDP (SQGDP) tables, in chained (inflation-adjusted) 2017 dollars to avoid conflating real output changes with inflation (Bureau of Economic Analysis, n.d.). We compute quarter-over-quarter growth rates (in percentage changes) and hold each quarter's growth rate constant across its three constituent months, since the underlying data is quarterly. For this analysis, we chose to go with growth rates rather than GDP levels because level variations are largely captured by other factors of our model: state fixed effects absorb cross-sectional level differences, and cycle fixed effects absorb cycle-level means. Moreover, growth rates are thought to have a more salient effect on consumer (and investor) economic sentiment, which might drive demand in the housing market.

2.2.4 State Population

Annual mid-year state population estimates from the U.S. Census Bureau (n.d.) are used as regression weights rather than as a regressor. The underlying survey is conducted yearly across all states, and population data is available from the 1900s to present. Instead of using the same value for the 12 months of a year, we chose to linearly interpolate from annual to monthly frequency from one annual data point to the next, effectively assuming that the population of every state grows by a constant amount each month between data releases. We believe that population grows in a sufficiently smooth and predictable fashion that linear interpolation introduces negligible error, and since population enters our models as a weight, small deviations versus actual (but unavailable) values are expected to have minimal impact on our coefficient estimates.

2.3 Sample Construction

2.3.1 Cycles and Event Time

To organize our panel around the electoral calendar, we define cycles: each May-to-April block constitutes one cycle, named by the year containing November. For example, May 2012 through April 2013 constitutes cycle 2012. This construction avoids splitting the election-to-inauguration window across calendar year boundaries, keeping the full November-through-April period - encompassing the election, the transition, and the initial months of the new administration - within a single observational unit.

Within each cycle, we define an event-time variable τ that maps each month to its position relative to November: May (6 months before the election) is $\tau = -6$, June is $\tau = -5$, and so on through November ($\tau = 0$) and April of the following year ($\tau = 5$). Each cycle is classified as presidential (`is_pres = 1` for 2012, 2016, 2020, and 2024), midterm (`is_mid = 1` for 2014, 2018, and 2022), or non-election (all remaining odd-year cycles). The motivation for this construction and the choice of reference period are discussed in greater detail in our methodology section.

Table 1: Summary Statistics (Excluding 2020 Presidential Cycle)

Statistic	N	Mean	St. Dev.	Min	Median	Max
Pending Sales	7,777	6,832.97	8,382.84	1	4,057	55,548
ln(Pending Sales)	7,777	8.16	1.27	0.69	8.31	10.93
Mortgage Rate (%)	7,777	4.59	1.28	2.84	4.07	7.62
Unemployment Rate (%)	7,777	4.69	1.87	1.70	4.20	30.50
GDP Growth (%)	7,777	0.43	1.21	−15.23	0.52	5.17
State Population (thousands)	7,777	6,352.54	7,219.23	572.07	4,424.07	39,505.26

2.3.2 Excluding the 2020 Presidential Cycle

The 2020 presidential cycle is excluded from the primary analysis on substantive grounds. The COVID-19 pandemic, federal stimulus transfers, near-zero interest rates, and the structural shift toward remote work produced housing market dynamics with no precedent in other cycles – the mid-2020 surge in pending sales, for instance, is well-documented as pandemic-driven rather than electorally driven. Including a cycle whose dominant variation is attributable to a concurrent global shock would contaminate our estimates of interest regardless of the direction of any resulting bias. The 2020 cycle is retained in Section A.1.1, where its inclusion is shown to substantially widen confidence intervals and shift several pre-election coefficients – a pattern consistent with, though not required to justify, its exclusion here.

2.3.3 Summary Statistics

Table 1 presents descriptive statistics for the analysis sample. After excluding the 2020 presidential cycle, the dataset comprises 7777 complete observations across 51 states (including the District of Columbia) over 12 cycles, with three presidential, three midterm, and 6 non-election cycles providing the variation used in estimation. Some additional complete observations before May 2012 and after April 2025 are retained, and contribute to the non-election baseline for our model below.

3 Methodology

In this section, we develop our econometric specification step by step, building up from a simple model to our final design. Each subsection introduces a design consideration, explains why the naive approach is inadequate, and presents the corresponding improvement. The final equation emerges at the end as the product of these cumulative refinements.

3.1 Event-Time Construction and Cycles

A natural starting point for studying election effects is to define a variable measuring the temporal distance between each observation and the nearest election. In our initial models, the event-time variable was computed as the number of months to the closest presidential election. This creates a four-year cycle, with event-time varying from 24 months before an election to 23 months after one. However, this approach introduces two problems. First, the baseline becomes difficult to motivate: we are trying to find the excess effect of a general election on the seasonal, non-election trend. An indicator that factors into the model only when the observation is 20 months away does not prove to be informative for our purposes. Second, this design disallows us from being able to account for midterm elections.

To resolve these issues, we remap time using twelve-month cycles (May-to-April blocks), each assigned to exactly one election type or to no election. Within each cycle, τ denotes the month's position relative to November, running from $\tau = -6$ (May) to $\tau = 5$ (April). This construction ensures that every month has a unique event-time value within its cycle, that presidential and midterm cycles are cleanly separated, and that non-election cycles provide an explicit baseline. For instance, cycle 2012 (May 2012 through April 2013) is presidential, while cycle 2013 (May 2013 through April 2014) carries no election and contributes to the seasonal baseline.

3.2 Identification Strategy

With event time defined, the next question is what we ought to account for. Panel data contains variation along multiple dimensions - across states, across time, and across the seasonal cycle - and not all of this variation is useful for identifying the effects of elections on housing transaction volume. Fixed effects estimation addresses this by absorbing group-level means: it is equivalent to demeaning the data within each group, subtracting the group mean from every observation and estimating on the residual variation.

For our purposes, two sources of confounding are particularly relevant. First, states differ permanently in market size, geography, and regulatory environment - California most likely would have more pending home sales than Wyoming, regardless of which time period the two states might be in. By adding state fixed effects (α_s), we hope to absorb these time-invariant differences, allowing us to ask: within each state, how do housing outcomes change over time? Second, aggregate market conditions shift over time - all states might undergo a national shock at the same time (like the 2008 financial crisis), and have the same effect happen to their housing markets. Cycle fixed effects (γ_c), where each May-April block receives its own intercept, absorb these between-cycle differences. After demeaning by both state and cycle, what remains is how the *seasonal shape* within a given cycle differs from one cycle type to another - precisely the variation we need.

For readers more familiar with the statistics tradition, we briefly note that fixed effects and random effects are two approaches to handling group-level heterogeneity in panel data. Both produce similar point estimates as the number of observations per group grows, but fixed effects is the default in applied econometrics when the groups represent the full population of interest (all 50 states rather than a random sample) and when group-level characteristics may correlate with regressors.

With this intuition in place, consider a natural first specification - the pooled event study:

$$\ln(Y_{st}) = \sum_{\tau=-5}^{5} \beta_\tau \mathbf{1}[\tau] + \alpha_s + \alpha_m + \alpha_y + \varepsilon_{st}$$

where D_{st}^{τ} is an indicator for event-time τ, α_s are state fixed effects, α_m are month-of-year fixed effects, and α_y are year fixed effects. This specification suffers from two shortcomings.

First, the month-of-year fixed effects (α_m) average across both election and non-election years. The "January" fixed effect blends Januaries in which a presidential inauguration occurred with Januaries in which no such event took place. The event-time coefficients β_τ therefore capture deviations from a contaminated baseline rather than from a clean non-election seasonal norm.

Second, the year fixed effects (α_y) are misaligned with the treatment window. The election-to-inauguration period spans November deep into the new year, crossing a calendar year boundary. Since the inauguration marks the start of a new policy regime, having this transition coincide with a change in year fixed effects means that the estimated effect of an electoral January could be partially attributed to the new year's change in intercepts rather than to the inauguration itself.

Our difference-in-events interaction model resolves both issues:

$$\ln(Y_{st}) = \alpha_s + \gamma_c + \sum_{\tau=-5}^{5} \beta_\tau \mathbf{1}[\tau] + \sum_{\tau=-5}^{5} \delta_\tau (\mathbf{1}[\tau] \times \text{IsPres}_c) + \sum_{\tau=-5}^{5} \phi_\tau (\mathbf{1}[\tau] \times \text{IsMid}_c) + \varepsilon_{st}$$

The cycle fixed effects (γ_c) replace year fixed effects, aligning the temporal grouping with the treatment window. The event-time indicators $\mathbf{1}[\tau]$ replace month-of-year fixed effects, since within the cycle framework each τ maps uniquely to one calendar month. The omitted reference category is $\tau = -6$ (May), chosen because it is far enough from the November election to be plausibly unaffected by election-specific dynamics while still permitting us to observe whether pre-election trends emerge.

This new model also allows us to make more transparent interpretations: β_τ captures the seasonal baseline - how much a given month in the cycle differs from May in non-election years. δ_τ measures the presidential election excess: how much a presidential year further shifts pending sales at month τ beyond the baseline non-election seasonal trend. ϕ_τ captures the corresponding midterm deviation. It is worth noting that these roles are distinct: $\tau = -6$ anchors the numerical scale (all coefficients are measured as differences from May), while $\tau = 0$ anchors the interpretation, being the period in which the election actually happens, and after which is the definite resolution of uncertainty.

This structure is broadly analogous to a difference-in-differences design, where the "treatment" is a presidential election and the "control" is the non-election seasonal baseline. However, an important distinction applies: there is no cross-sectional control group, as all states undergo the same election simultaneously. In other words, rather than invoking parallel trends across groups, identification here rests on a different assumption. By drawing conclusions from this model, we are assuming that each state's seasonal pattern is stable across its non-election cycles, such that non-election years provide a valid counterfactual for what election years would have looked like absent the election.

3.3 Controls, Weighting, and Inference

3.3.1 Controls

Even after absorbing state and cycle means via fixed effects, and controlling for seasonality trends through β_τ, within-cycle macroeconomic variation could still confound or inflate the election effect. Mortgage rates fluctuate monthly with Federal Reserve decisions, unemployment varies with local labor market shocks, and GDP growth might affect consumer sentiment and behavior - all factors that affect housing demand independently of elections. We include three contemporaneous controls: the national 30-year mortgage rate, state-level unemployment, and state-level GDP growth. A robustness check with lagged controls (Section A.1.3) produces similar effects, suggesting that the controls capture genuine confounds rather than absorbing part of the election effect itself. Adding macro controls somewhat attenuates excess effect estimates but does not eliminate the overall pattern; a progressive-controls analysis is presented in Section A.1.2.

3.3.2 Weighting

For this analysis, we weight observations by state population so that our estimates reflect the nationally representative effect.

To motivate this choice, consider a simplified four-state panel: Wyoming, Alaska, Texas, and California. Under unweighted OLS, each state receives an equal $\frac{1}{4}$ share of influence on the final estimates - despite California having roughly 67 times the population of Wyoming, and Texas having roughly 42 times the population of Alaska. This assigns a disproportionately large role to small states whose housing markets, while important locally, represent a small fraction of national buyer activity. Population-weighted OLS corrects for this: assigning weights proportional to each state's population, the two largest states (California and Texas) together account for roughly 98% of the estimator's weight, while Wyoming and Alaska contribute the remaining 2%. The resulting estimates reflect the election-cycle pattern as it would be experienced by the typical American home buyer, rather than the typical state.

Formally, population-weighted OLS minimizes $\sum_i w_i(y_i - \hat{y}_i)^2$ where w_i is proportional to state population, and can be implemented rather painlessly by specifying relevant arguments in modern regression estimation packages.

3.3.3 Standard Error Clustering and Inference

Standard OLS assumes independent errors, but panel data routinely violates this: shocks persist within states over time (serial correlation) and Federal Reserve announcements affect all states simultaneously (cross-sectional correlation). Clustered standard errors address this by allowing errors within the same cluster to be arbitrarily correlated; full derivations of the cluster-robust variance estimator from first principles are presented in Cameron and Miller (2015).

A useful way to quantify the problem is through the variance inflation factor: for a coefficient $\hat{\beta}_k$, the ratio of the true clustered variance to the default OLS variance is approximately

$$\tau_k \approx 1 + \rho_{x_k} \rho_u (\bar{N}_g - 1),$$

where ρ_{x_k} is the within-cluster correlation of regressor k, ρ_u is the within-cluster error correlation, and $\bar{N}_g$ is the average cluster size (Cameron and Miller, 2015). In our setting, national-level regressors like mortgage rates ($\rho_{x_k} = 1$ between states in the same month) pair with big clusters, meaning the penalty for ignoring clustering can be substantial - an example in the paper showed that a standard error inflation factor $\sqrt{\tau_k} > 3$ is not unusual in similar designs even when correlation terms can be small.

A single clustering dimension handles correlation along one axis but not both. Cameron and Miller (2015) extend the framework to multi-way clustering via a variance matrix that combines one-way estimates across dimensions and subtracts the double-counted intersection:

$$\widehat{V}_{\text{2way}}[\widehat{\beta}] = \widehat{V}_{\text{state}}[\widehat{\beta}] + \widehat{V}_{\text{time}}[\widehat{\beta}] - \widehat{V}_{\text{state}\cap\text{time}}[\widehat{\beta}].$$

The intuition is additive: two-way clustering asks whether residuals are correlated within states *and* within time periods, and removes the overlap to avoid double-counting. The method relies on the number of clusters per dimension, so it performs well when both dimensions have many clusters - a condition our state and calendar-month dimensions comfortably meet. The same inclusion-exclusion logic applies to our conservative three-way specification, which adds cycle as a third clustering dimension: under three-way clustering, errors within a given presidential cycle are also allowed to be correlated, which is the natural assumption if aggregate national conditions shift in ways that affect all states jointly during a given electoral period.

We estimate our model under two clustering specifications:

1. **State and time (calendar-month)**: This is our primary specification.
2. **State, time, and cycle**: Our more conservative specification, which adds cycle-level clustering.

Under the primary specification, several estimates are statistically significant; under the conservative specification, significance vanishes. This is a predictable consequence of the data structure: with only approximately 12 full cycles and only 3 classified as presidential, the effective number of clusters along the cycle dimension is extremely small. We present results under both and interpret accordingly.

3.4 Final Specification and Hypotheses

Bringing together each of the components described above, our final specification is:

$$\ln(Y_{st}) = \alpha_s + \gamma_c + \sum_{\tau=-5}^{5} \beta_\tau \mathbf{1}[\tau] + \sum_{\tau=-5}^{5} \delta_\tau (\mathbf{1}[\tau] \times \text{IsPres}_c) + \sum_{\tau=-5}^{5} \phi_\tau (\mathbf{1}[\tau] \times \text{IsMid}_c) + \mathbf{X}'_{st}\lambda + \varepsilon_{st}$$

where $\ln(Y_{st})$ is the natural log of pending home sales for state s in month t, α_s are state fixed effects, γ_c are cycle fixed effects, $\mathbf{1}[\tau]$ are event-time indicators with $\tau = -6$ omitted, IsPres_c and IsMid_c are cycle-type indicators (whose main effects are absorbed by the cycle fixed effects), and $\mathbf{X}_{st}$ is a vector of macroeconomic controls (mortgage rate, unemployment rate, and GDP growth). Observations are weighted by state population, and errors are clustered as described in the section above.

Based on Bernanke's (1983) framework and the empirical literature reviewed in Section 1, we present the following hypotheses:

- **H1 (Pre-election friction):** The presidential excess δ_τ should be negative for $\tau \in \{-5, \dots, -1\}$. Election uncertainty depresses pending sales below the seasonal baseline as households exercise their option value of waiting, consistent with Bernanke's model.
- **H2 (Post-election release):** δ_τ should be positive for $\tau \in \{0, 1\}$ and beyond. The resolution of electoral uncertainty triggers a burst in transactions as pent-up demand re-enters the market.

- **H3 (Post-resolution reversion):** δ_τ should return toward zero as τ increases. As the new administration takes office and its policy priorities become clearer, the market normalizes.
- **H4 (Presidential versus midterm):** Presidential and midterm elections should produce differing effects on housing markets, given the differences in their stakes, policy scope, and the presence or absence of a full regime transition. We expect the midterm excess effects (ϕ_τ) to be smaller in magnitude than the presidential estimates, but to remain statistically discernible from the baseline seasonal trend.

3.5 A Note on the Estimation Framework

We estimate all models using `fixest::feols()` in R, which implements concentrated-out (also known as demeaning) fixed effects estimation for multi-way fixed effects models. This is the standard approach in applied microeconomics for panel data with many groups.

An alternative from the statistics tradition would be mixed-effects models (e.g., `lme4` in R) or Bayesian hierarchical models (e.g., `brms`), which treat group intercepts as random draws from a distribution and estimate both group-level and residual variance simultaneously. These approaches offer advantages in small-sample settings through partial pooling - relevant given that our analysis rests on only three presidential cycles. However, key inference techniques used in our analysis - particularly multi-way standard error clustering and population weighting - have well-developed implementations in the fixed-effects framework but limited counterparts in standard mixed-effects software. Small-sample refinements such as wild bootstrap standard errors are available in `fixest` but might be more naturally motivated in the mixed-effects context, where the hierarchical structure could accommodate the small number of cycles more directly. Unfortunately, this would require substantial additional specification and theoretical motivation beyond the scope of this paper.

We proceed with the fixed-effects approach, acknowledging that the small effective sample size for cycle-level inference is a binding limitation regardless of our estimation strategy.

4 Results

4.1 Main Results

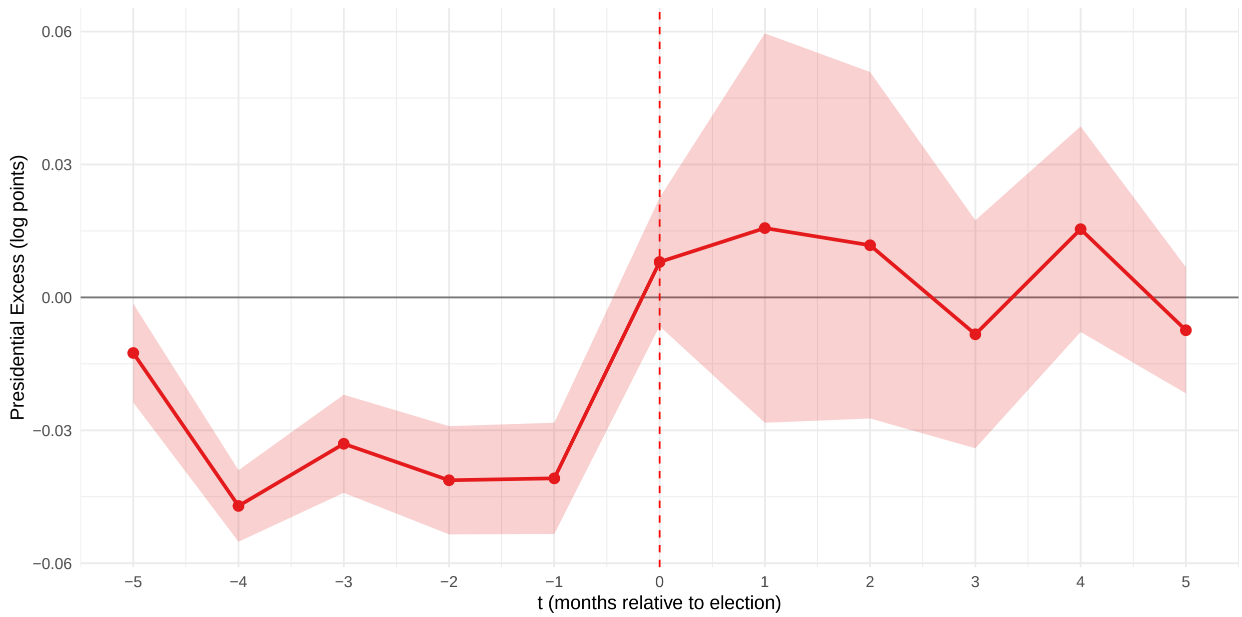

Figure 1: Presidential Excess on ln(Pending Sales). State + time clustering, pop-weighted, excluding 2020. Shaded band denotes 95% confidence interval.

Figure 1 presents the presidential excess coefficients (δ_τ) from our preferred specification, estimated with state and time clustering, population weighting, and the full set of macroeconomic controls.

The pre-election period ($\tau = -5$ through $\tau = -1$) shows consistent negative excess, supporting H1. Pending sales in presidential election years run approximately 3 to 5 percent below the non-election seasonal baseline throughout this window, with the suppression staying at -4.1% the month before the election. This is consistent with households exercising their option value of waiting: faced with policy uncertainty that will only resolve at the ballot box, buyers defer commitments to irreversible housing purchases.

On the other hand, H2 is not supported by our results. At $\tau = 0$ and $\tau = 1$ (November and December), the presidential excess turns positive but small - at about 1% higher than the baseline seasonal trend - and is statistically indistinguishable from zero, even under our more lenient clustering setup. In other words, the anticipated burst of pent-up demand does not materialize as a detectable spike; rather, housing activity simply ceases to be suppressed, becoming indistinguishable from non-election intertemporal housing trends.

In the post-inauguration period ($\tau = 2$ through $\tau = 5$), coefficients oscillate near zero with no systematic pattern, consistent with H3: the market normalizes to its seasonal baseline after enough time from the resolution of uncertainty.

Table 2: Presidential and Midterm Excess Coefficients Under Primary and Conservative Clustering

	Preferred Spec		Conservative Spec	
τ	Presidential	Midterm	Presidential	Midterm
-5	-0.013*	-0.013	-0.013	-0.013
	(0.006)	(0.023)	(0.029)	(0.020)
-4	-0.047***	-0.047*	-0.047+	-0.047
	(0.004)	(0.024)	(0.029)	(0.033)
-3	-0.033***	-0.026	-0.033	-0.026
	(0.006)	(0.022)	(0.043)	(0.028)
-2	-0.041***	-0.023	-0.041	-0.023
	(0.006)	(0.027)	(0.034)	(0.052)
-1	-0.041***	-0.008	-0.041	-0.008
	(0.006)	(0.021)	(0.039)	(0.046)
0	0.008	-0.044*	0.008	-0.044
	(0.007)	(0.021)	(0.040)	(0.036)
1	0.016	-0.054	0.016	-0.054
	(0.022)	(0.050)	(0.054)	(0.081)
2	0.012	-0.062+	0.012	-0.062*
	(0.020)	(0.033)	(0.034)	(0.030)
3	-0.008	-0.080**	-0.008	-0.080**
	(0.013)	(0.031)	(0.028)	(0.030)
4	0.015	-0.050+	0.015	-0.050
	(0.012)	(0.027)	(0.031)	(0.041)
5	-0.007	-0.046	-0.007	-0.046
	(0.007)	(0.029)	(0.054)	(0.063)
Mortgage Rate	-0.083***		-0.083***	
	(0.011)		(0.016)	
Unemp. Rate	-0.005		-0.005	
	(0.017)		(0.017)	
GDP Growth	0.029*		0.029*	
	(0.014)		(0.014)	
N	7777		7777	
Within R^2	0.287		0.287	
Within Adj. R^2	0.284		0.284	

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2 reports the full set of regression coefficients under both clustering specifications. Under the primary specification, the pre-election coefficients ($\tau = -4$ through $\tau = -1$)

range from -0.033 to -0.047 and are statistically significant at the 1% level or better. Post-election coefficients ($\tau = 0$ through $\tau = 5$) are uniformly small and insignificant, ranging from -0.008 to $+0.016$. Among the macro controls, mortgage rate has the largest and most precisely estimated effect (-0.083, $p < 0.001$), consistent with its central role in housing affordability: unsurprisingly, higher mortgage rates result in lower transaction volumes. Additionally, GDP growth rates remain significant across both clustering specifications (0.029), but only at the 10% level.

It is also worth noting that in the preferred clustering specification, the standard error estimate for all excess effect terms increase substantially after the election. Beyond simple statistical artifacts, this can also be a sign of heterogeneity in post-election market response across states after the election.

4.2 Conservative Clustering

Under the conservative specification - which adds cycle as a third clustering dimension - the same pre-election coefficients lose statistical significance. Across the board, the addition of cycle clustering increased SE estimates anywhere from 1.5x to 5x compared to the preferred specification. The point estimates remain directionally consistent, but the confidence intervals widen considerably, and individual coefficients can no longer be distinguished from zero at conventional significance levels.

This attenuation is a predictable consequence of the data structure. With only 12 cycles in the sample and only 3 classified as presidential, the effective sample size for cycle-level inference is extremely thin. Clustering by cycle essentially asks: across the few independent election cycles we observe, are these deviations systematic? With three treated cycles, the capacity to reject the null hypothesis is inherently limited.

Hence, we interpret this as a suggestive finding: the pre-election friction documented under primary clustering is consistent with theoretical predictions, but cannot be considered definitive given the small effective sample size.

4.3 Presidential Elections vs. Midterms

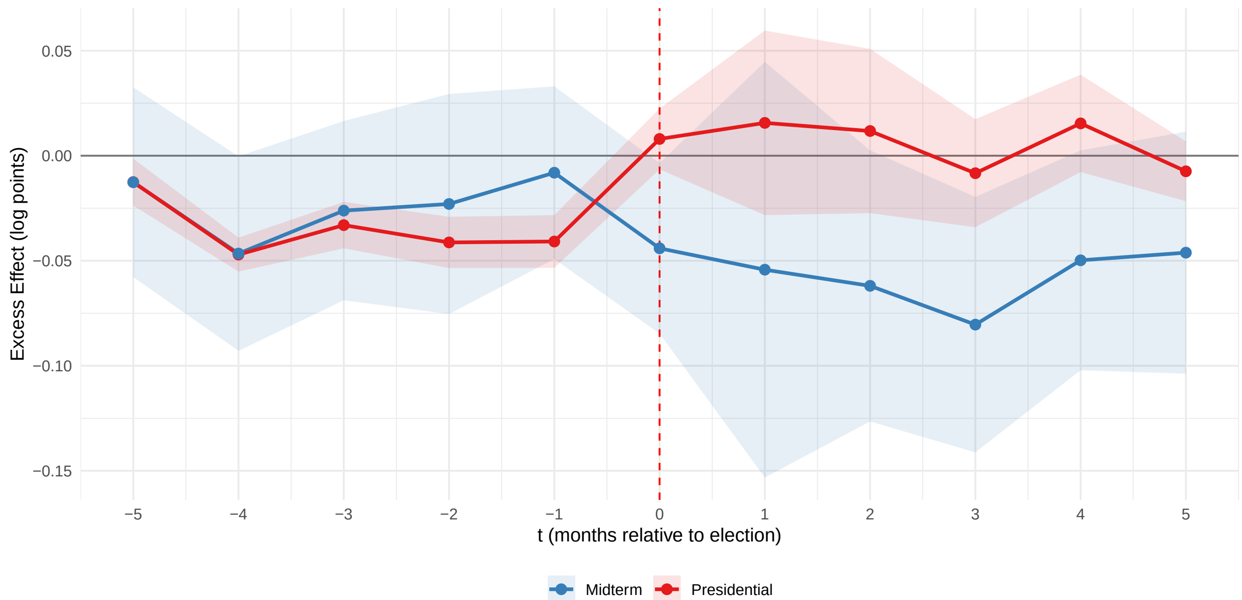

Figure 2: Presidential versus Midterm Excess on ln(Pending Sales). Full controls, state + calendar month clustering, population-weighted, excluding 2020. Shaded bands denote 95% confidence intervals.

Figure 2 overlays the presidential and midterm excess coefficients estimated under the same preferred specification. The midterm estimates (ϕ_τ) stay close to zero and are statistically indistinguishable from the non-election baseline throughout the cycle. This contrasts sharply with the presidential cycle pattern and supports a revised H4: while presidential cycles do exert a depressive effect on home sales leading up to the election, the excess effects of midterm cycles are indistinguishable from normal seasonality.

Additionally, this comparison serves as an informal placebo check. If the presidential pattern were driven by a generic even-year effect - for instance, a biennial cycle in media coverage or consumer sentiment unrelated to elections - midterm excess effects should show a similar pattern. The fact that midterm cycles are nigh-indistinguishable to the baseline strengthens the interpretation that the presidential pattern is linked to features specific to presidential elections: uncertainty over federal policy direction, and the inauguration serving as a salient termination of the current economic status quo.

5 Discussion and Conclusion

The timing of the observed pattern is partially consistent with Bernanke's (1983) option value framework. Agents face uncertainty about future policy conditions throughout the election season, exercise their option to wait, and transact after the election resolves the uncertainty. The post-inauguration reversion is consistent with the market returning to its normal seasonal trajectory once the policy environment stabilizes under the new administration.

It is worth briefly comparing our results with Nguyen and Vergara-Alert (2023), who found that gubernatorial elections reduce annual housing transactions by 2,000 to 3,200 per state. Our monthly decomposition reveals a temporal pattern - a pre-election dip followed by a post-election return to normalcy - that may aggregate to the same net effect seen in their estimates. We note, however, that these are different election types (gubernatorial versus presidential), use different data sources at different granularities, and employ different identification strategies, so direct comparisons should always be made with caution.

Our methodological contribution is the difference-in-events interaction design, which decomposes monthly housing dynamics into a non-election seasonal baseline, a presidential cycle effect, and a midterm cycle effect. This provides an explicit counterfactual - the seasonal pattern in non-election years - against which election-year deviations can be measured, addressing a limitation of prior pooled event study approaches that relied on contaminated baselines and misaligned fixed effects.

Limitations temper the strength of our conclusions. While the exogeneity of election timing provides a credible baseline for causal interpretation, observing only three presidential cycles yields a very thin basis for confident causal claims. Furthermore, because all states experience the election simultaneously, there is no within-year variation in treatment status that would allow us to use "untreated" states as a concurrent control group. The sensitivity to clustering confirms this fragility: under more conservative inference that accounts for cycle-level correlation, individual coefficients lose statistical significance.

Despite these limitations, our findings demonstrate that the electoral cycle does have a real effect on economic activity beyond regular macroeconomic policy channels, opening clear avenues for future work. Two directions for future research appear particularly promising. First, swing-state heterogeneity: if election uncertainty is the operative channel, the excess election effect should concentrate in states where electoral outcomes are more uncertain, or even differ between states that are "competitive" versus states that are not. This is partly indicated by the results we outlined above: after the election, our standard error estimates increases substantially, hinting at response heterogeneity across the sample. Second, property-type heterogeneity: the single-family home market might be compositionally different to the market for condominiums and other commercial properties. Disaggregating the analysis by property type would test this prediction directly. Ultimately, as higher-frequency datasets are made available and more advanced statistical frameworks continue to mature, future research will be well-positioned to refine our understanding of exactly how, and for whom, elections might affect decision-making.

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A Appendices

A.1 Robustness Checks

A.1.1 Including the 2020 Presidential Cycle

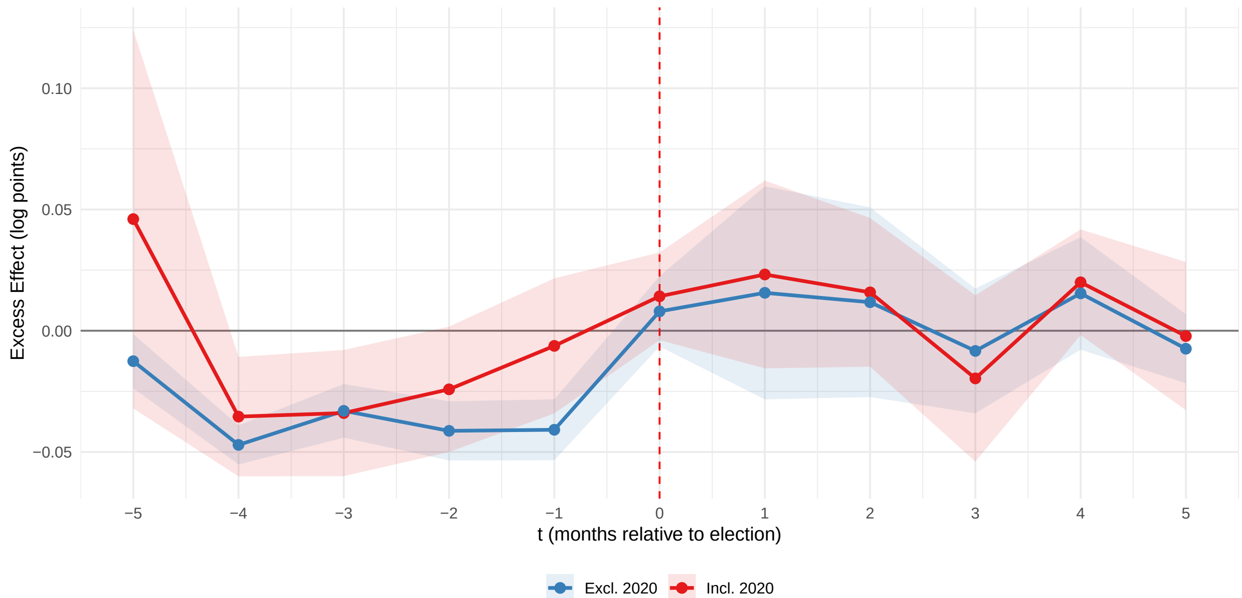

Figure 3: Presidential Excess: Including versus Excluding the 2020 Cycle. Same specification as the main event study.

Re-running the preferred specification with the 2020 presidential cycle included produces notably different estimates. The confidence bands widen substantially for the pre-election period, reflecting the increased variance introduced by the pandemic-distorted cycle. Some coefficients shift in unexpected directions - for instance, the June estimate ($\tau = -5$) is now positive and significant (in no small part due to the massive rise in home sales in Mid 2020) - and the majority of coefficients that were significant under the primary specification lose significance. This sensitivity to a single anomalous cycle underscores the fragility inherent in estimating election effects from a small number of national events, and supports the exclusion of 2020 from the main analysis.

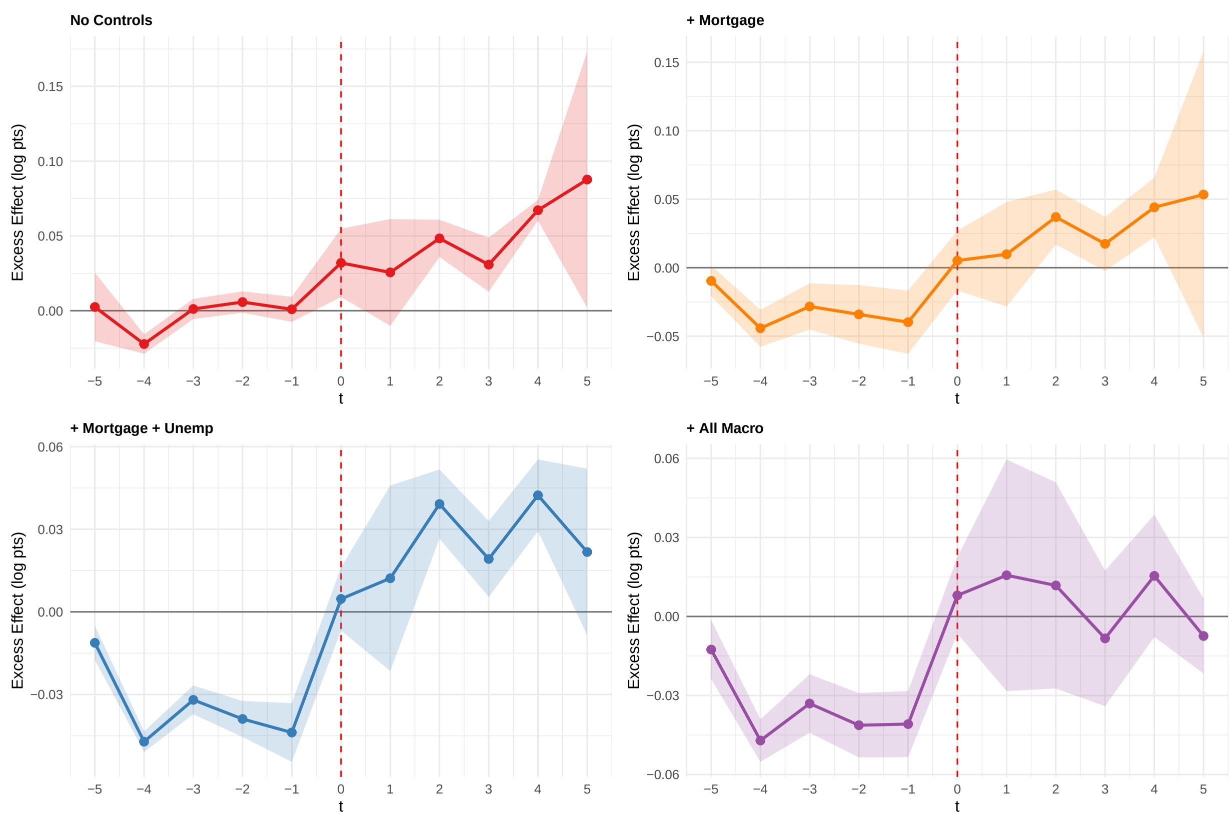

Figure 4: Progressive Controls: Presidential Excess Under Successive Additions of Macro Controls. All panels use state + calendar month clustering, population-weighted, excluding 2020.

A.1.2 Progressive Macro Controls

We sequentially add macroeconomic controls to the no-controls baseline to examine how each variable affects the presidential excess effect. Without macroeconomic controls, the coefficients seem to hover around the zero line before the election, and are positive and significant after the election. However, just the addition of mortgage rates alone shifted the pre-election effect downwards, and the addition of other macroeconomic pointers only made the effect more pronounced. This implies that the initial null finding was most likely driven by differences in macroeconomic outcomes across cycles, rather than the net effect of general elections, and that the true effect will likely be closer to what our most stringent specification is estimating.

A.1.3 Lagged Macroeconomic Controls

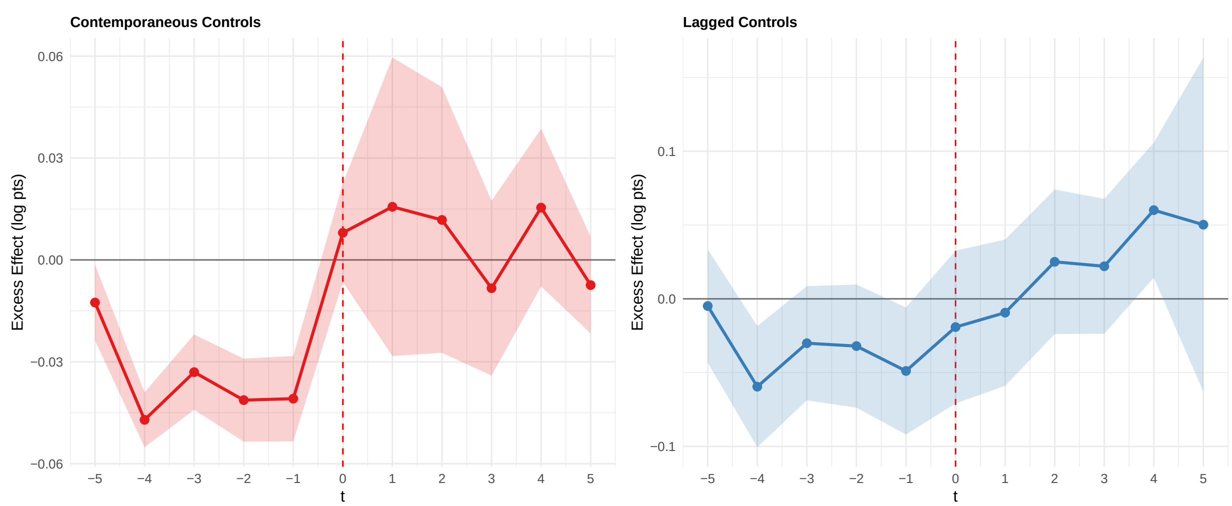

Figure 5: Contemporaneous versus Lagged Macro Controls. Both use state + calendar month clustering, population-weighted, excluding 2020.

We re-estimate the event-study using lagged macro controls (unemployment, GDP growth, mortgage rates) in place of contemporaneous values to address potential causal concerns. The qualitative pattern remains similar – estimates hover in the negatives pre-election and turn positive after Election Day – though standard errors widen for all periods and the post-election profile becomes more gradual. This sensitivity is consistent with macro variables (especially interest rates) reacting quickly to election outcomes, so the contemporaneous specification should be interpreted as the election's excess effect net of macroeconomic conditions, while the lagged-control specification provides a conservative check on timing and sign.

A.2 AI Use

This project made use of an AI coding agent runnning on Anthropic's Claude Opus 4.6 model (accessed through the Google Antigravity IDE platform) throughout multiple stages of the research process. The author takes full responsibility for all content, claims, and any errors in the final work. No AI-generated code or writing used without author oversight.

Coding and analysis. The AI agent was responsible for a substantial portion of the executable code in the final report. Its most significant contribution was in producing the results section's graphics - both the LaTeX-formatted coefficient table (Table 2) and the coefficients-comparison figures (Figure 1, Figure 2, Figure 3, Figure 4, Figure 5) - where custom helper functions were created to extract and plot model output from the difference-in-events model whose structure did not integrate well with standard event-study graphing packages. The agent was also used to explore additional analyses (e.g., alternative outcome variables) that were ultimately not included. For data preparation, the collection process was manual, but cleaning and merging was carried out by the agent under extensive monitoring and correction by the author. The agent proved unreliable at correctly merging heterogeneous data files and required repeated prompting to ask clarifying questions about data structure before proceeding. The final analysis dataset and the code that produced it will be included in the final submission.

Formatting. LaTeX formatting code was initially written manually by the author and subsequently verified by the agent for correct specification.

Writing and ideation. The background and literature review was adapted from a prior course project (ECON 481) and expanded manually by the author; the agent was used for initial feedback but its suggestions were largely unrelated to the analysis and were rejected. The data section was drafted by the author using a framework from the same prior project, with the agent following the provided structure; overstatements and inaccuracies were corrected by the author, and the motivation for variable inclusion and the exclusion of the 2020 cycle were the author's own. For the results and discussion sections, model output was initially summarized by the agent, but all interpretations, claims, and commentary in the final paper were written by the author. The discussion section followed a similar pattern to the background - the agent's contributions were not substantively helpful.

Proofreading. The agent was valuable in detecting misspellings, grammatical errors, and suggesting alternative phrasing, but was largely uninfluential in shaping main ideas or substantive claims.